

PART 1: Company Profile

- **Company Name:** ABC Startup Solutions
- **Nature of Business:** Software Development
- **Company Vision:** To be the leading provider of innovative, scalable software solutions that empower global businesses to thrive in the digital economy.
- **Company Mission:** To deliver high-quality, secure, and user-centric software applications while fostering a culture of continuous technological advancement and operational excellence.
- **Office Location:** 123 Tech Avenue, Innovation District, San Pablo City, Laguna, Philippines
- **Organizational Structure:** Flat Hierarchy (Executive Management → Department Heads → Team Members)
- **Employee Distribution:**
 - Information Technology: 5
 - Human Resources: 4
 - Finance: 5
 - Sales: 6
 - **TOTAL:** 20

PART 2: Enterprise Hardware Inventory

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Laptops (High-Spec)	5	Information Technology	Mobile workstations for software development and administrative tasks.
HW-002	Desktop Computers	15	HR, Finance, Sales	Standard workstations for daily office productivity, CRM, and accounting.

Asset ID	Hardware	Quantity	Department	Purpose
HW-003	Enterprise Server	1	Shared / IT	Hosts company databases, internal applications, and central directory.
HW-004	Gigabit Router	1	Shared / IT	Routes traffic between the local network and the internet.
HW-005	24-Port PoE Switch	1	Shared / IT	Connects all wired office devices to the local network.
HW-006	Network Printer	2	Shared	Centralized document printing and scanning (1 for IT/Sales, 1 for HR/Finance).
HW-007	UPS (1500VA)	20	All Departments	Provides temporary battery backup for desktops and networking gear during power outages.
HW-008	Wireless Access Point	2	Shared / IT	Provides Wi-Fi coverage for the entire office floor.
HW-009	NAS Storage	1	Shared / IT	Centralized file sharing and automated local data backups.
HW-010	External Backup Drive	2	IT	Offline, air-gapped backup storage for disaster recovery.

Asset ID	Hardware	Quantity	Department	Purpose
HW-011	24" IPS Monitors	20	All Departments	Primary and secondary displays for employee workstations.

PART 3: Enterprise Software Inventory

Software	Version	License	Purpose
Windows 11 Pro	23H2	Commercial	Primary Operating System for employee workstations, providing domain join and BitLocker.
Ubuntu Server	24.04 LTS	Open Source	Secure, stable Operating System for the main company server.
Microsoft Office	Microsoft 365	Subscription	Standard productivity suite for document creation, spreadsheets, and emails.
VS Code	Latest	Open Source	Primary integrated development environment (IDE) for the IT development team.
Git	Latest	Open Source	Version control system for tracking software code changes.
GitHub Desktop	Latest	Open Source	Graphical user interface to simplify Git version control workflows.
VirtualBox	7.0	Open Source	Hypervisor for creating isolated virtual machines to test software deployments.

Software	Version	License	Purpose
Google Chrome	Latest	Freeware	Standardized web browser for accessing cloud-based tools and internet resources.
Microsoft Defender	Built-in	Commercial	Endpoint security to protect systems against malware and viruses.
AnyDesk	Latest	Commercial	Enables the IT team to provide remote desktop support to end-users.
7-Zip	Latest	Open Source	Utility for compressing and extracting archive files to save storage space.

PART 4: Enterprise Network Inventory

Equipment	Quantity	Purpose
ISP Modem	1	Demarcation point connecting the company to the Internet Service Provider.
Router	1	Manages IP routing and acts as the gateway between the WAN and LAN.
Hardware Firewall	1	Inspects incoming and outgoing traffic to block unauthorized access and cyber threats.
24-Port Switch	1	Core distribution point linking all internal wired devices (servers, PCs, printers).
Wireless Access Point	2	Broadcasts the corporate SSID for wireless laptops and mobile devices.

Equipment	Quantity	Purpose
24-Port Patch Panel	1	Terminates Ethernet cables from wall ports for clean server rack management.
CAT6 Cables	2 Boxes	High-speed Ethernet cabling for wiring the office layout (1000ft per box).
RJ45 Connectors	100 pcs	Connectors used to terminate the ends of the CAT6 networking cables.

PART 5: Enterprise Network Diagram

(Note: Use the structural logic below to create your final export in Draw.io. Ensure you include the Diagram Title and your Student Name on the canvas).

Code snippet

graph TD

Internet((Internet)) --> Modem[ISP Modem]

Modem --> Router[Enterprise Router]

Router --> Firewall[Hardware Firewall]

Firewall --> CoreSwitch[24-Port Switch]

CoreSwitch --> Server[(Ubuntu Server)]

CoreSwitch --> WAP[Wireless Access Point]

CoreSwitch --> Printer[Network Printer]

CoreSwitch --> IT[IT Department: 5 Laptops]

CoreSwitch --> HR[HR Department: 4 Desktops]

CoreSwitch --> Finance[Finance Department: 5 Desktops]

CoreSwitch --> Sales[Sales Department: 6 Desktops]

WAP -> WiFiDevices[Wireless Devices]

PART 6: System Administration Roles

Helpdesk Technician

- **Responsibilities:** Acts as the first point of contact for user IT issues, troubleshoots hardware/software problems, resets passwords, and configures workstations.
- **Skills:** Strong customer service, basic networking, OS troubleshooting (Windows/Mac), active listening.
- **Common Tools:** Ticketing systems (Jira, Zendesk), AnyDesk, Active Directory Users and Computers, diagnostic utilities.
- **Certifications:** CompTIA A+, Microsoft 365 Certified: Endpoint Administrator.

Network Administrator

- **Responsibilities:** Designs, deploys, and maintains the physical and logical network infrastructure (LAN/WAN), manages firewalls, and monitors network performance.
- **Skills:** Subnetting, routing protocols, firewall configuration, network security, packet analysis.
- **Common Tools:** Wireshark, Cisco Packet Tracer, SolarWinds, PuTTY.
- **Certifications:** Cisco CCNA, CompTIA Network+.

Linux System Administrator

- **Responsibilities:** Manages enterprise Linux servers, configures web/database hosting environments, automates server tasks, and ensures system security and uptime.
- **Skills:** Command-line proficiency, bash scripting, server hardening, storage management, virtualization.
- **Common Tools:** SSH, cron, Vim/Nano, systemd, Apache/Nginx.
- **Certifications:** CompTIA Linux+, Red Hat Certified System Administrator (RHCSA).

Cloud Administrator

- **Responsibilities:** Migrates local infrastructure to cloud environments, manages cloud computing resources, ensures cloud security compliance, and monitors cloud billing.

- **Skills:** Cloud architecture, identity and access management (IAM), containerization, disaster recovery planning.
- **Common Tools:** AWS Management Console, Azure Portal, Terraform, Docker.
- **Certifications:** AWS Certified SysOps Administrator, Microsoft Certified: Azure Administrator Associate.

How They Work Together:

Inside one organization, these four professionals form a cohesive IT lifecycle team. The Network Administrator builds the foundational infrastructure that allows systems to communicate securely. The Linux and Cloud Administrators deploy and manage the critical backend servers and hosted applications across this network. Finally, the Helpdesk Technician supports the end-users who access these systems daily, escalating complex network outages or server failures back to the specialized administrators, ensuring seamless business operations from the backend hardware to the end-user interface.

PART 7: Infrastructure Recommendations

- **Internet Provider:** Procure a dedicated fiber-optic enterprise connection with a minimum of 500 Mbps symmetrical speeds and a Service Level Agreement (SLA) for 99.9% uptime to support development workflows.
- **Server Specifications:** The central server should utilize a modern 16-core processor, at least 64GB of ECC RAM, and RAID 10 NVMe SSD storage. This ensures fault tolerance and sufficient resources for local virtualization and database hosting.
- **Backup Strategy:** Implement the 3-2-1 backup rule. Maintain three total copies of data, across two different storage mediums (NAS and External Drives), with one copy kept off-site or securely in the cloud to prevent total data loss during physical disasters.
- **Security Recommendations:** Implement a dedicated hardware firewall with Intrusion Detection/Prevention Systems (IDS/IPS). Separate the internal network into VLANs (e.g., isolating Finance from IT and Guest WiFi) to minimize lateral movement in case of a breach.
- **Antivirus:** Deploy a centrally managed Endpoint Detection and Response (EDR) solution rather than standalone consumer antivirus, allowing IT to monitor all 20 endpoints from a single dashboard.
- **Password Policy:** Enforce a minimum 12-character passphrase requirement, mandate Multi-Factor Authentication (MFA) for all email and server access, and implement a 90-day password rotation policy for administrative accounts.

- **Expansion Plan:** Purchase a 24-port switch instead of a 16-port switch to accommodate immediate growth without hardware replacement. Design the server rack with additional U-space for future server nodes as the developer team expands.

- **PART 8: Personal Reflection**

Through this enterprise infrastructure planning project, I learned the comprehensive process of designing an IT environment from the ground up. I realized that system administration extends far beyond troubleshooting broken hardware; it requires deep strategic alignment between business objectives and technology solutions. Translating the theoretical needs of a startup into a concrete, realistic inventory of hardware, software, and networking components highlighted the meticulous attention to detail required in this profession, especially when considering budget, security, and specific departmental needs. The most challenging task was designing the enterprise network topology and selecting realistic hardware configurations for the inventory. Balancing the operational needs of different departments—such as ensuring the IT department has high-performance development tools while Finance and Sales require reliable daily productivity setups—required careful thought and research. Furthermore, logically mapping out the network with proper security layers, from the ISP modem to the hardware firewall and the internal distribution switches, tested my understanding of network architecture and traffic flow. Planning is absolutely critical before deployment because it prevents costly mistakes and ensures scalability. Purchasing equipment without a solid blueprint often leads to hardware incompatibilities, security vulnerabilities, and network bottlenecks that disrupt business operations. A well-documented infrastructure plan serves as a roadmap that guides physical installation, simplifies future troubleshooting, and aligns IT investments with the company's growth trajectory. Without it, the network would become chaotic, insecure, and unmanageable as the startup scales. This project will significantly help me become a better System Administrator by shifting my mindset from reactive problem-solving to proactive architecture design. I now understand the importance of creating professional technical documentation, communicating technology needs to business stakeholders, and applying security at every layer of deployment. By mastering these foundational planning phases, I am better equipped to build resilient, secure, and efficient IT environments that can support the continuous operations of any organization I join in the future.